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12.2 IT Technical Support and Management

Mohammed Abdulwasa Mohammed Qasem

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Task 1 - Introduction

As staff member working for a small PC repair shop in the centre of Coventry. The company has started running the outsourced support service and several customers have been already signed up.

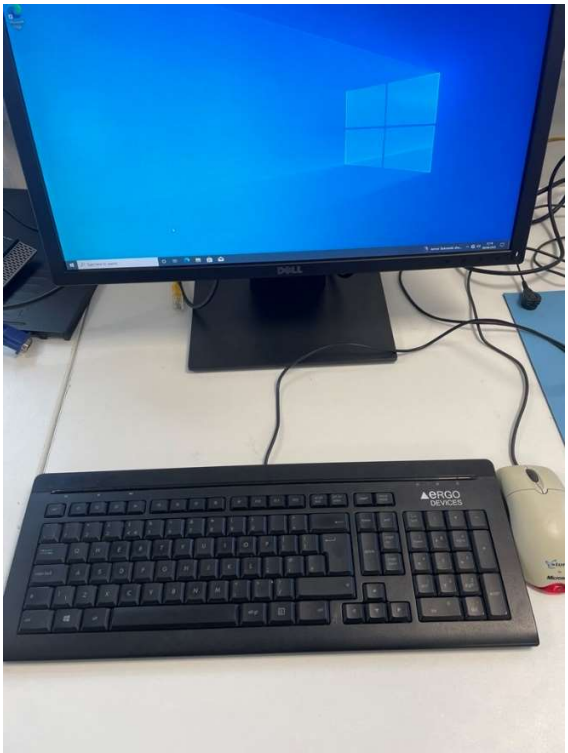
In this section of the report, we will carefully carryout at least 6 IT support and management tasks safely. For example:

- Peripheral installations and configurations
- Hard drive installation
- Software upgrade
- Faulty user issues
- Disk configuration
- Faulty in accessing the internet.

Routine 1/Activity - Peripherals Installations and Configurations

In the first routine/activity we have a monitor, mouse and keyboard peripherals that needs to be checked if they function properly as intended, the first step I did was checking all the peripherals are connected to the PC, second step was checking if I can prompt anything such as moving the mouse around, typing the keyboard specifically each letter and symbol of the keyboard and for the monitor we will only be checking whether there's delay or nothing is moving/performed.

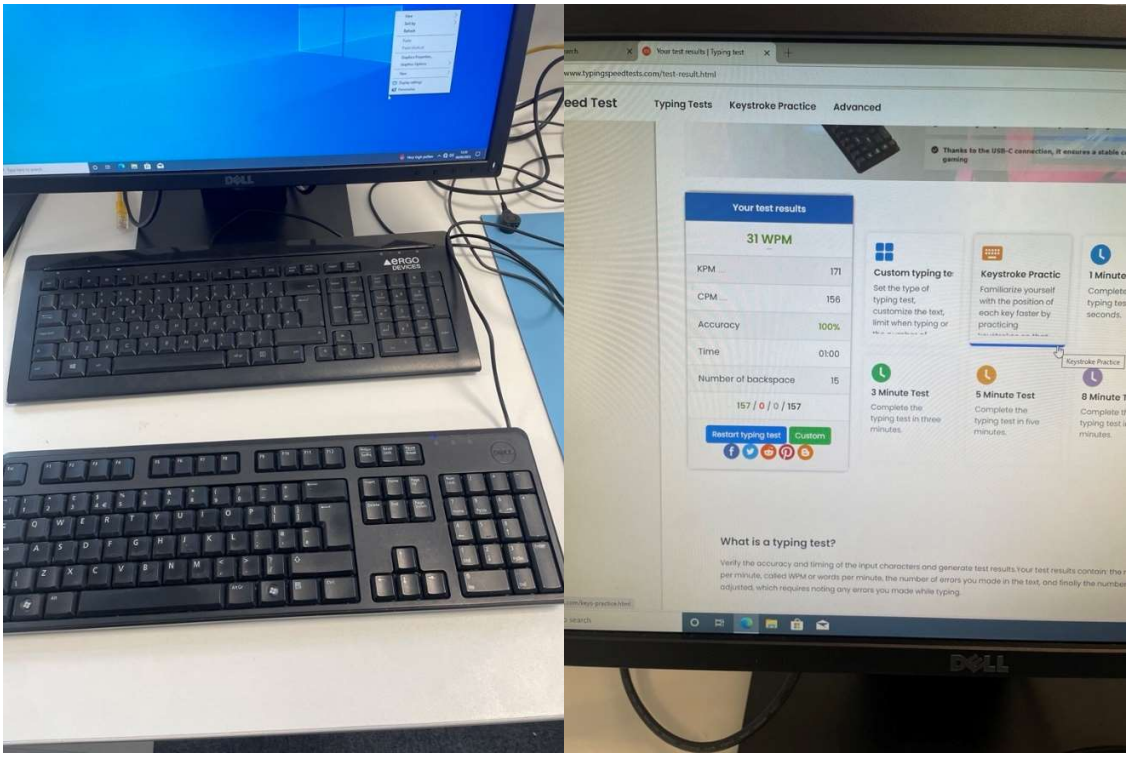
Below is picture evidence showing that the monitor responds to mouse movement. However, we observed that none of the keyboard keys are functioning.



Possible Solution

To fix this issue, firstly we need to double check if it is not only the issue with the keyboard and potentially could be drive or virus in PC that makes the keyboard prompts unresponsive, which the issue is with the keyboard only. To do this we will have to bring another keyboard and run a test on website called (Typing Club) which to emphasise that each key in the keyboard working as intended.

As we suspected, the issue was with the original keyboard, where certain keys such as O, E, Q, D, A, and 6 were unresponsive. After testing with a replacement keyboard, all keys functioned correctly without any issues.



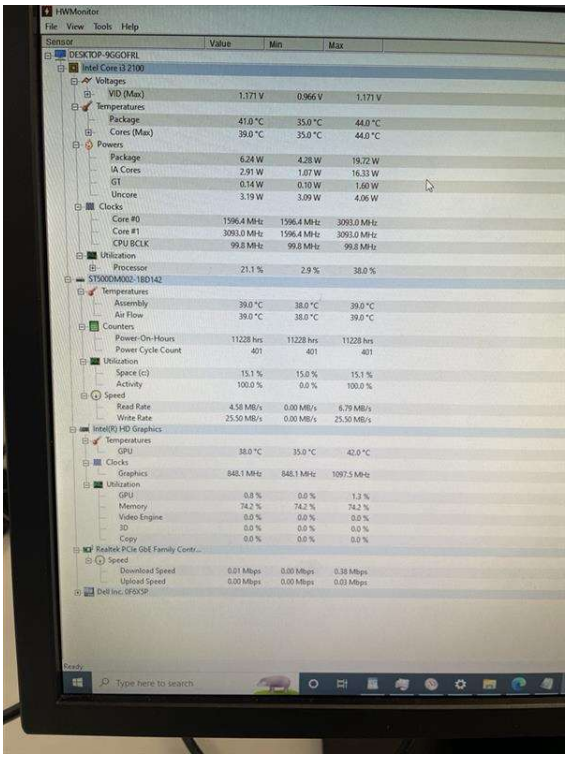
Routine 2/Activity - Hard Drive Cooling System Installation

The second activity we will be checking one of the PCs that we have if the Hard Drive is updated and configured properly, since its owner been telling us that the hard drive was overheating sometimes which leads to low performance of the PC. Below the screenshot evidence of the hard drive and Task manager performance.

Potential problems that make the hard drive overheat could be, the PC do not have proper cooling systems, dust build-up, or physical placements like placing it to another heat source component like CPU. To inspect these issues, we will dismantle the PC and see it closely.

Possible Solution

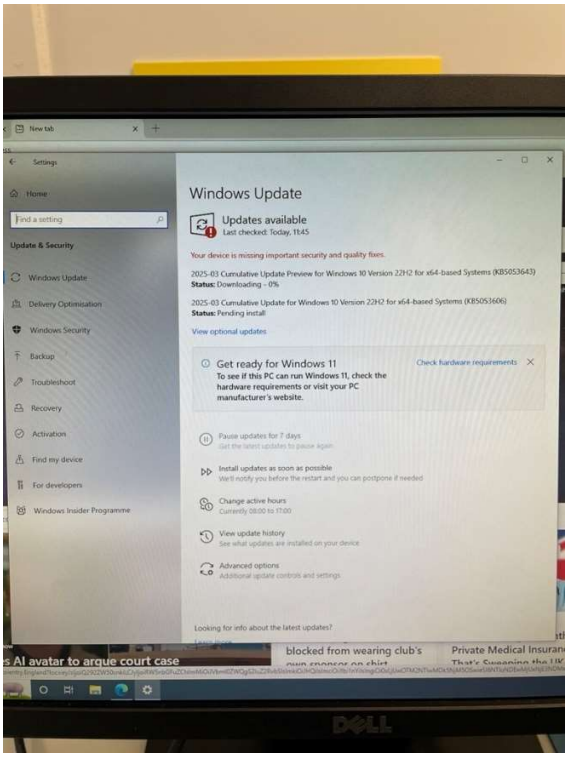
To see any difference change after installing cooling systems, we will run HWMonitor app to see the temperature, fan speed, power etc, on the PC. Below is the evidence.



The result of the test shows a normal temperature of CPU and all other components which what we implemented was effective in cooling down the computer.

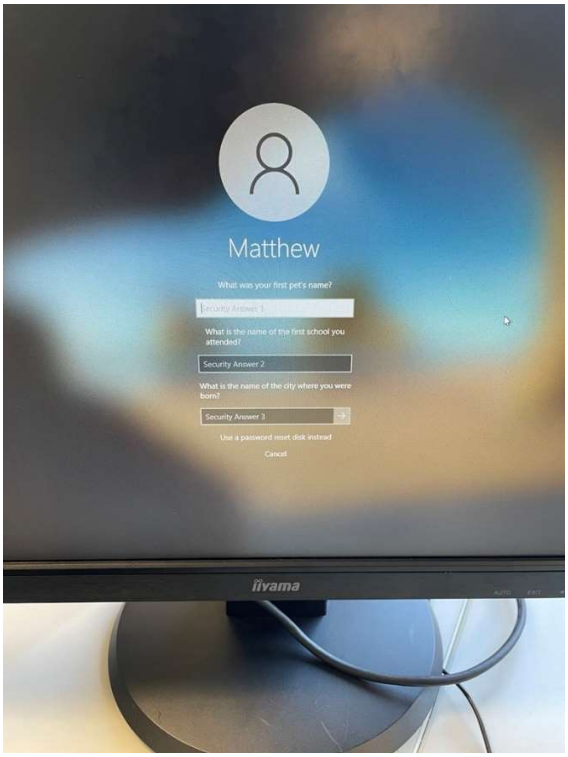
Routine 3/Activity - Software Upgrade

While I was configuring one of the PCs, I found out that it needs an important security and quality fixes through installing an update for the device. This update will ensure to keep the device more secure and matches with the latest encryption features. Below is the evidence of the installation.



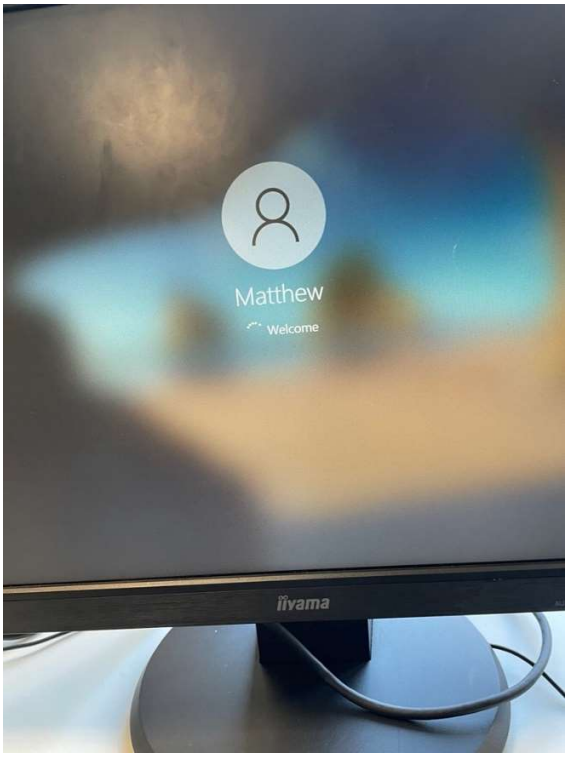
Routine 4/Activity - Faulty Login User Issues

One of the students in Coventry College has logging issues because he forgotten his password, I asked the student if he remembers any security questions, the student was not able to provide me with those security answers also he forgotten them. Below is the user logging issue.



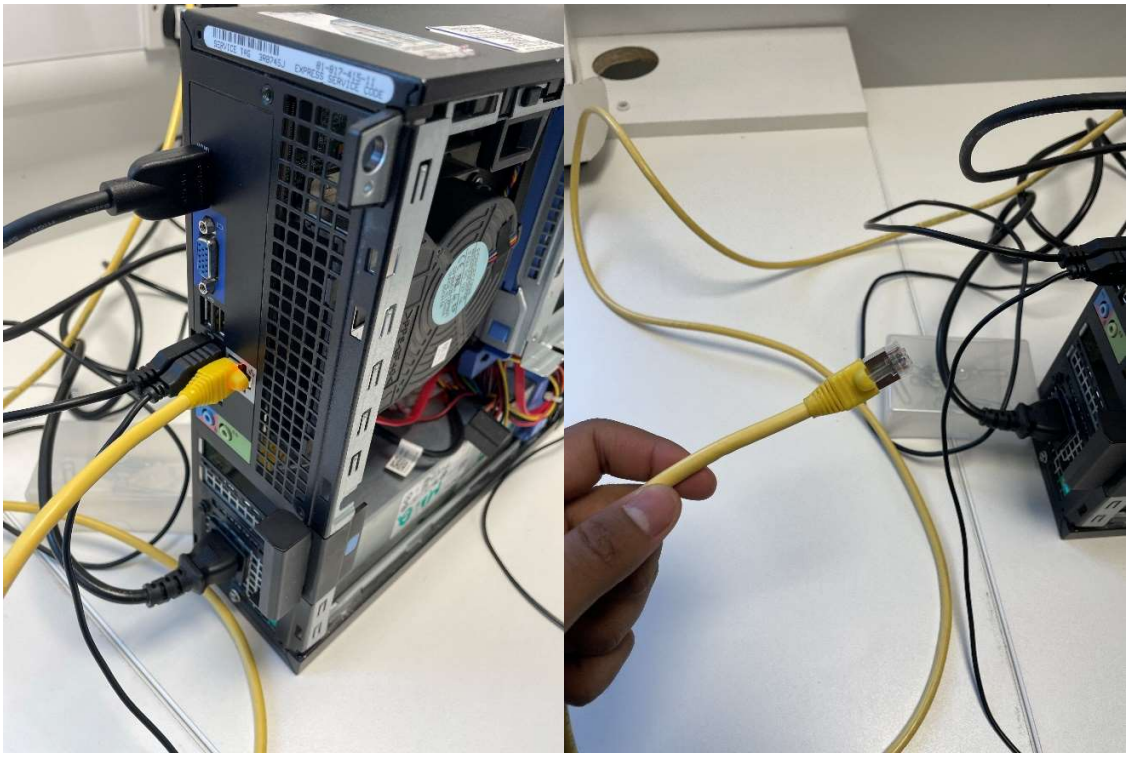
Possible Solution

To fix this issue, I had to look up for the organisation security file archives and find the security questions regarding the student who previously forgotten his password and those security answers. I have managed to find them and try them out to logging, and it worked. Below is the evidence.



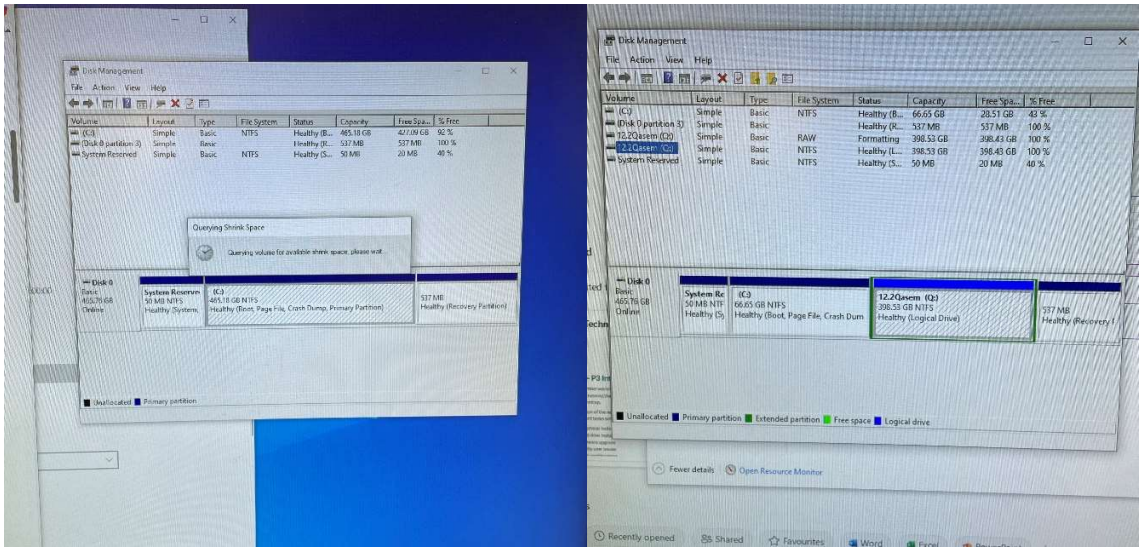
Routine 5/Activity - Fault Accessing the Internet

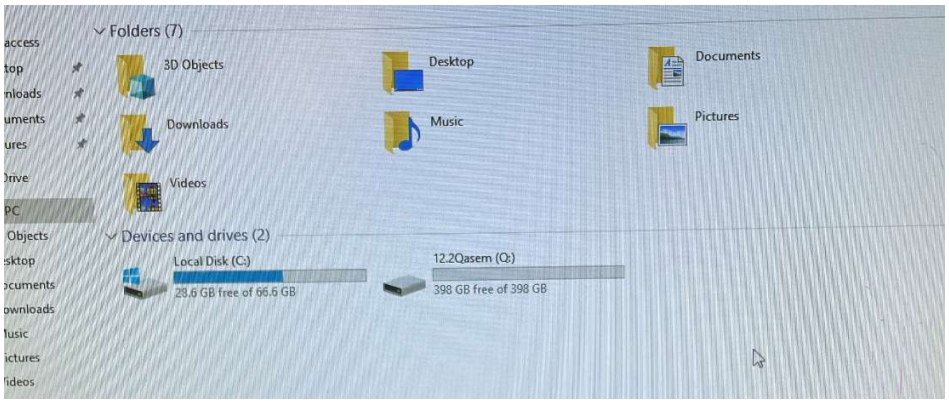
The Lab room where many PCs are there cannot access the internet because of the low Wi-Fi coverage frequency, to fix this problem we managed to install ethernet cables on the wall and handed each computer specific ethernet cable to be plugged into each PC to allow internet access in that computer. Below is the evidence of the ethernet cable?



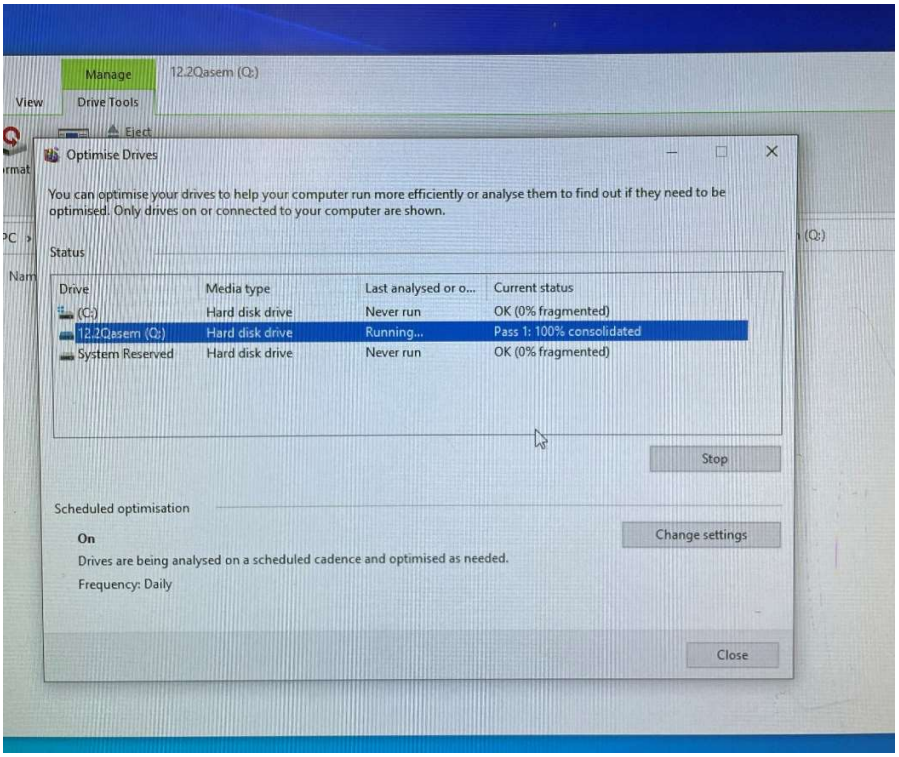
Routine 6/Activity - Disk Configuration

To configure the disk, firstly we need to shrink a copy of the main disk such as in this case the C: disk we used as shrink and we named the shrink as 12.2 Qasem. Below is the evidence of these steps.

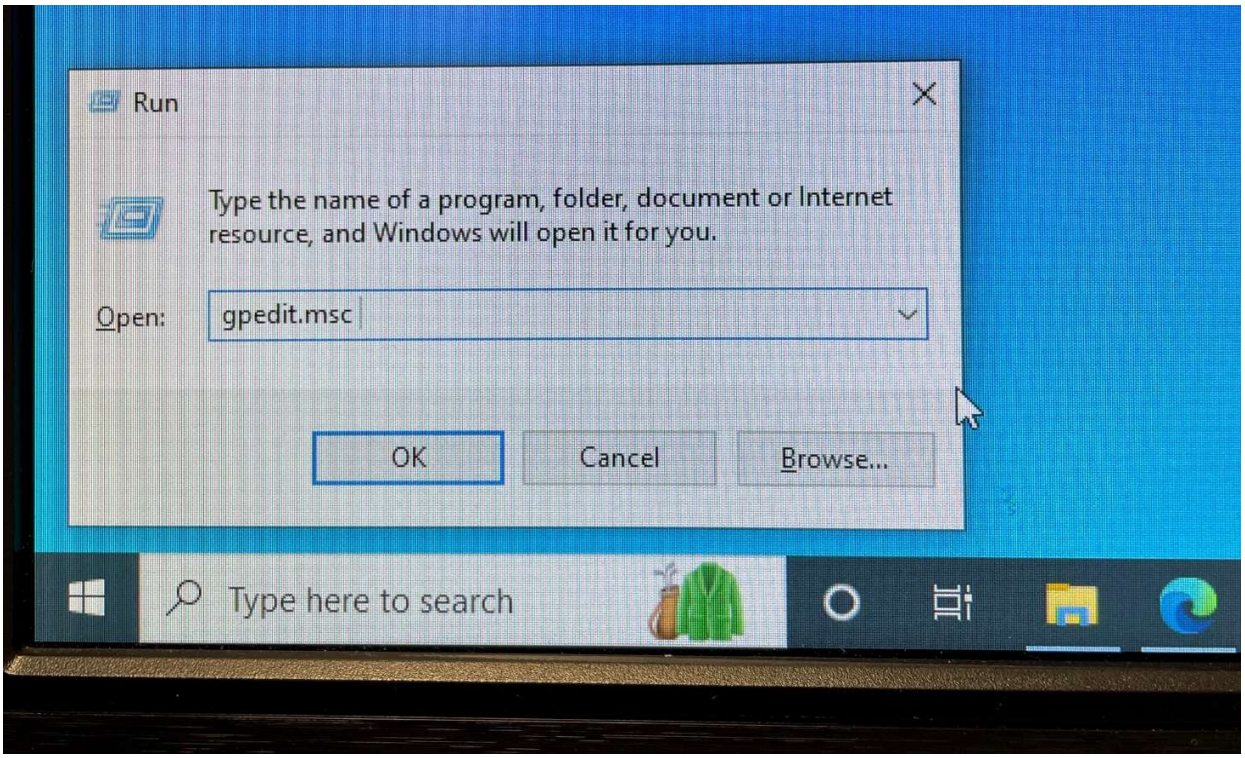




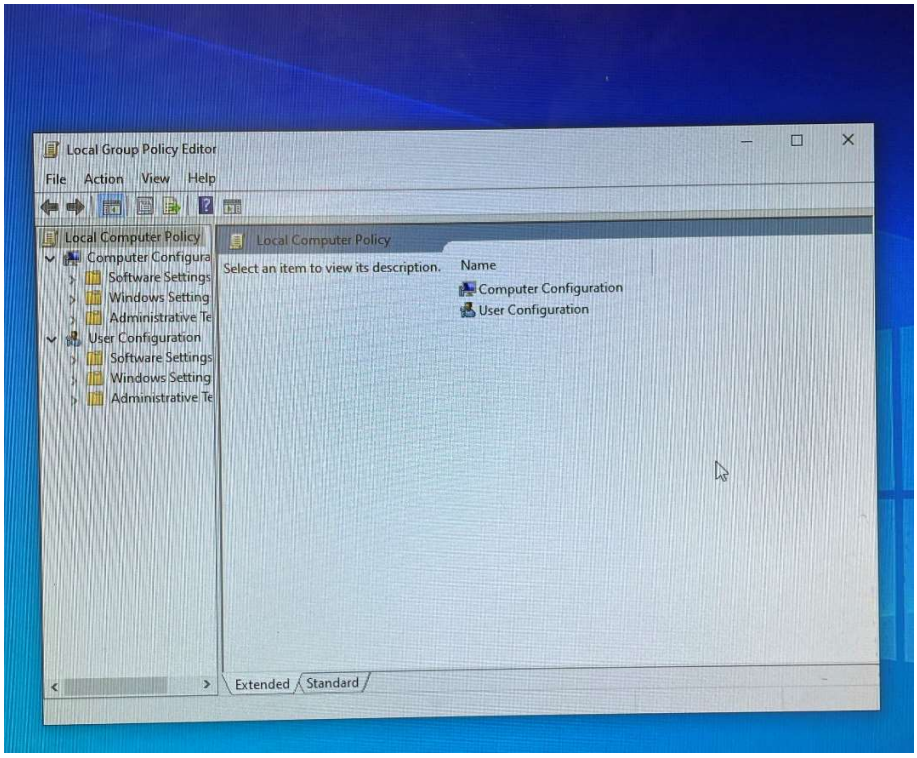
Another step we did was optimising this new disk to ensure the drive help the computer to run efficiently or analyse them to find out if they need to be optimised. Below is the evidence of this step.



Final procedure we managed to do is creating a group policy for the new disk to see if it works properly. Below is the prompt for opening the local group policy editor.

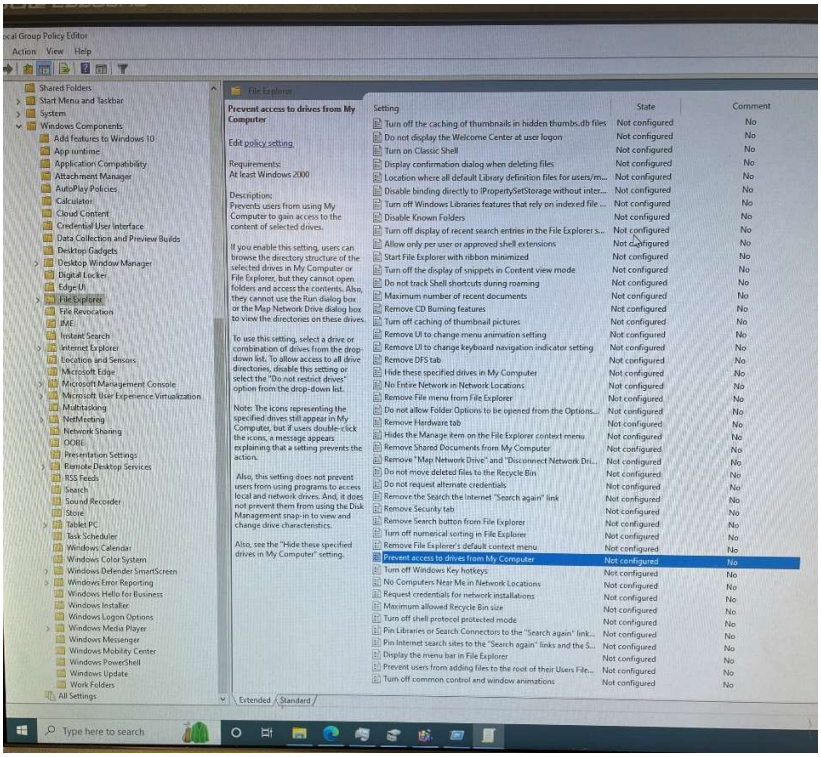


This prompt allows the user to open this tab. Below is the evidence.

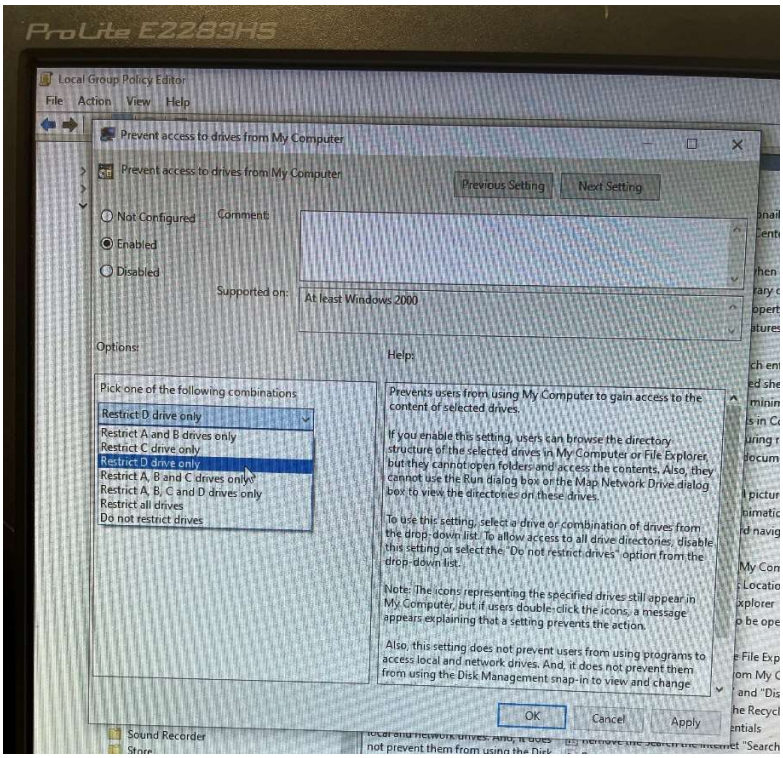


This editor page allows users to add any group policy but for this case we will only explain the steps we did to implement the (prevent access to drives from my computer) this policy enhance the security by restricting user access to local drives which prevents accidental or malicious data loss, modification or malware. The guide step by step:

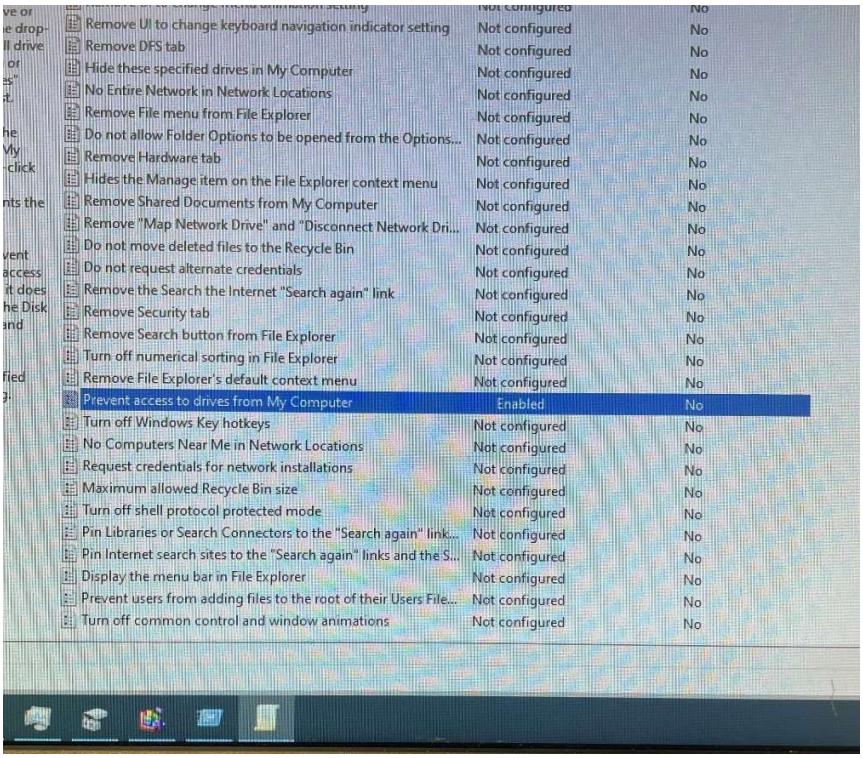
User Configuration> Administrative Templates> Windows Components> File Explore



In the File Explore we double clicked on the **Prevent access to drives from my computer** then this page has popped up:



After choosing to enable and from the open menu we chose restrict D drive only, below is the evidence of the policy we implemented being enabled and active.



Remove UI to change keyboard navigation indicator setting	Not configured	No
Remove DFS tab	Not configured	No
Hide these specified drives in My Computer	Not configured	No
No Entire Network in Network Locations	Not configured	No
Remove File menu from File Explorer	Not configured	No
Do not allow Folder Options to be opened from the Options...	Not configured	No
Remove Hardware tab	Not configured	No
Hides the Manage item on the File Explorer context menu	Not configured	No
Remove Shared Documents from My Computer	Not configured	No
Remove "Map Network Drive" and "Disconnect Network Dri...	Not configured	No
Do not move deleted files to the Recycle Bin	Not configured	No
Do not request alternate credentials	Not configured	No
Remove the Search the Internet "Search again" link	Not configured	No
Remove Security tab	Not configured	No
Remove Search button from File Explorer	Not configured	No
Turn off numerical sorting in File Explorer	Not configured	No
Remove File Explorer's default context menu	Not configured	No
Prevent access to drives from My Computer	Enabled	No
Turn off Windows Key hotkeys	Not configured	No
No Computers Near Me in Network Locations	Not configured	No
Request credentials for network installations	Not configured	No
Maximum allowed Recycle Bin size	Not configured	No
Turn off shell protocol protected mode	Not configured	No
Pin Libraries or Search Connectors to the "Search again" link...	Not configured	No
Pin Internet search sites to the "Search again" links and the S...	Not configured	No
Display the menu bar in File Explorer	Not configured	No
Prevent users from adding files to the root of their Users File...	Not configured	No
Turn off common control and window animations	Not configured	No

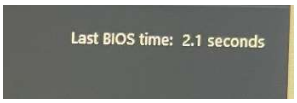
Monitoring performance

To monitor and optimise the performance of the system which can ensure it meets the client requirements for a secure system with adequate performance we will be using Task manager to configure the system in terms of tracking the last BIOS time, disabling some startup apps to minimise the BIOS time and we will reboot the system to check the final BIOS time. While doing that we will also, ensure to carry out these steps safely by wearing an anti-ESD wristband and having any hardware components on an anti-ESD mat.

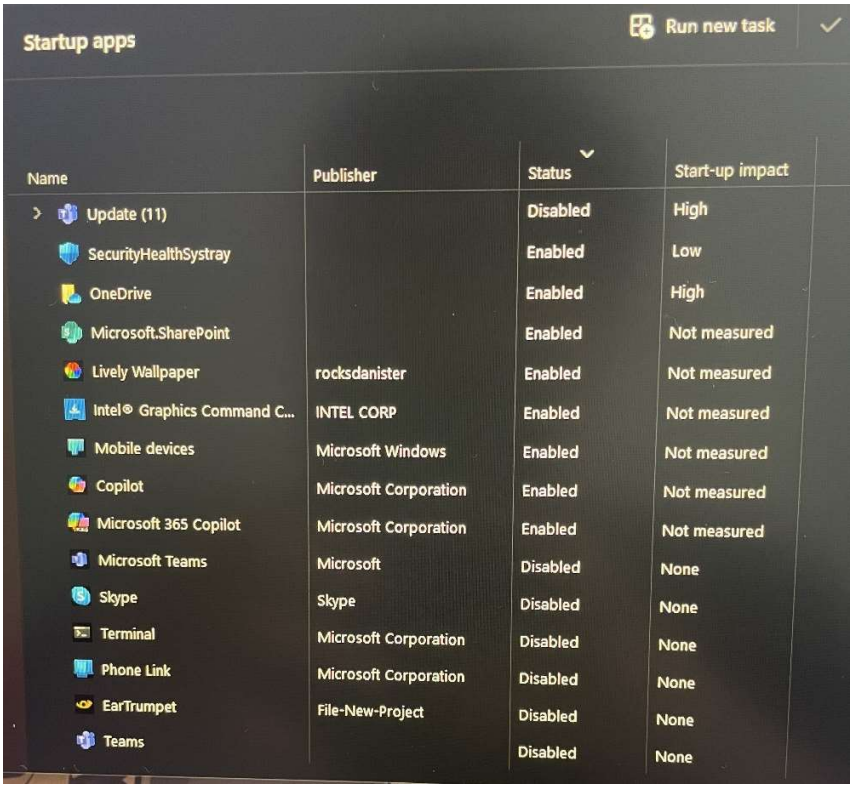
Additionally, we will back up the computer before completing a maintenance task. Below is all the evidence of these steps:

Anti-ESD wristband and anti-ESD mat

Before disabling the startup apps (last BIOS time)



Disabling startup apps



Startup apps

Run new task

Name	Publisher	Status	Start-up impact
Update (11)		Disabled	High
SecurityHealthSystray		Enabled	Low
OneDrive		Enabled	High
Microsoft.SharePoint		Enabled	Not measured
Lively Wallpaper	rocksdanister	Enabled	Not measured
Intel® Graphics Command C...	INTEL CORP	Enabled	Not measured
Mobile devices	Microsoft Windows	Enabled	Not measured
Copilot	Microsoft Corporation	Enabled	Not measured
Microsoft 365 Copilot	Microsoft Corporation	Enabled	Not measured
Microsoft Teams	Microsoft	Disabled	None
Skype	Skype	Disabled	None
Terminal	Microsoft Corporation	Disabled	None
Phone Link	Microsoft Corporation	Disabled	None
EarTrumpet	File-New-Project	Disabled	None
Teams		Disabled	None

After disabling startup apps (last BIOS time)

As we can see from this evidence, the initial BIOS time was 2.1 seconds due to many unnecessary startup apps that led to this delay which can cause slow apps rendering or the computer being lagging. To fix this issue we disable many unnecessary startup apps that is taking long time to render/start by disabling them and only enabling some apps that we think is essential to be used as startup apps, this way we managed to reduce the last BIOS time from 2.1 to 1.7 seconds which significantly is good for optimising performance.

Overview of Task 1 and Time/Behaviours Management

After carrying out six different IT solving problems, we managed to fully fix all the issues and documenting them while also, considering safety behaviours such as doing backup before altering or deleting anything to ensure we don't lose any files and data, wearing an anti-ESD wristband and having any hardware components on an anti-ESD mat helped to increase the safety of these procedures.

We managed to provide in-house support in the shop and out-house support for our clients such as going to Coventry college and fixing the user logging issue and fault in accessing internet. We managed to safely fix all the issues that our clients had faced demonstrating our technical knowledge and expertise. Additionally, we demonstrate our leadership and taking responsibilities by ensuring that we fix the client issue on agreed time and in case of any further issues we caused we will ensure that we take full responsibility of our actions and what we led to by apologising to the client and paying any compensation for any damages we caused.

Another key soft skill we managed to appropriately demonstrate was communicating with our clients and team members by using software tools and apps like Teams and Gantt chart to log and track our progress of any tasks that has been assigned to us. Below is the evidence of Gantt chart that we have used throughout these activities.

Task Title	31/03/2025	1/4/2025	2/4/2025	2/4/2025	2/4/2025	2/4/2025	2/4/2025	2/4/2025
Initiating the report								
Routine 1/Activity - Peripherals Installations and Configurations								
Routine 2/Activity - Hard Drive Cooling System Installation								
Routine 3/Activity - Software Upgrade								
Routine 4/Activity - Faulty Login User Issues								
Routine 5/Activity - Faulty in Accessing the Internet								
Routine 6/Activity - Disk Configuration								
P4 Monitoring performance								
Overview of Task 1 and Time/Behaviours Management								
Report closer for Task 1								
Done								
On Progress								
Not Started Yet								

Task 2 – IT Support and Management Plan

Abstract

This section of the report is support and management plan has been created for a start-up online business that employees between 25 and 30 people. The systems need to be available 24/7 - 365 days a year. All employees will use desktop PCs while the senior staff will have additional Wi-Fi connected tablet devices. There will be two shared printers available for staff to use. This plan will outline the company's approach to incident response, disaster recovery, capacity planning, sustainability and environmental waste planning, and preventive maintenance which will ensure that the IT infrastructure will be robust, efficient, and scalable.

Incident Response

The business will focus on minimising the downtime and maintaining operational continuity. As helpdesk system such as Freshdesk or ServiceNow will be implemented to log all incidents. The incidents will be prioritised based on the severity such as complete system outages which will require a response within 15 minutes while low priority issues like password reset will be responded to within 4 hours. If a major issue was not addressed within the time set, the business needs to escalate to level 2 or level 3 support within one hour. Any updates need to be communicated to affected users via internal email or Teams.

Disaster Recovery

Disaster recovery is essential part of the IT support strategy which is aimed to restore the critical services quickly in the event of failure. The full daily backups of all servers and workstations will be performed with critical business data backed up on an hourly basis. The data will be replicated in real time to ensure secondary location using cloud services

such as Azure backup. Full disaster recovery tests will be conducted twice annually to validate the effectiveness of the recovery plan.

Capacity Planning

The capacity planning will ensure the IT infrastructure can handle the current workforce and anticipated growth. Initially 30 high-performance desktop PCs will be deployed, each equipped with 16GB RAM, Intel i7 CPU and SSD for optimal performance. The senior staff will also be provided with tablet connected to the network via Wi-Fi. The network infrastructure will be around a Gigabit LAN supported by enterprise grade Wi-Fi 6 access points from manufacturers such as Ubiquiti UniFi. A dedicated server will be used to manage file sharing, authentication, and printing services, while customers facing platforms will be hosted in the cloud to ensure scalability and uptime.

Sustainability and Environmental Waste Planning

Sustainability will be integrated into all aspects of IT operations to reduce the environmental impact. All hardware's will be selected carefully based on energy efficiency ratings. Devices will be refreshed every five years, with older equipment responsibly recycled through certified e-waste partners. Smart power strips and optimised cooling systems will be used to reduce energy consumption. The printer will be defaulted to double sided printing to lower paper use. Additionally, the company will encourage remote and virtual meetings to minimise the environmental impact associated with business travel.

Additional Aspect: Preventive Maintenance

This will be an initiative-taking approach to avoid unexpected downtime and hardware failures. Such as implementing schedule monthly maintenance windows will allow for software updates and security patches to be applied systematically. Full hardware inspections will be carried out quarterly to identify any potential issues early.

Office Layout Plan

Each workstation (PCs) will be provided with ethernet cabling (Cat6) routed under the floor where possible to maintain a clean and safe environment and Wi-Fi access point will be installed at middle top near the senior staff member desk to provide seamless coverage.

Plan Justification

Every step/action has been decided in terms of IT support and Management Plan which helps to balance the reliability, scalability, environmental responsibility, and business efficiency. The above structure shows the initial office layout plan to ensure incident response which needs to be urgent during problems that need to be rapid action to minimise downtime. Disaster recovery measures can guarantee that data loss is limited, and the business can be able to resume its operation within short window after critical event/issue. The capacity planning allows the company to grow without significant future IT

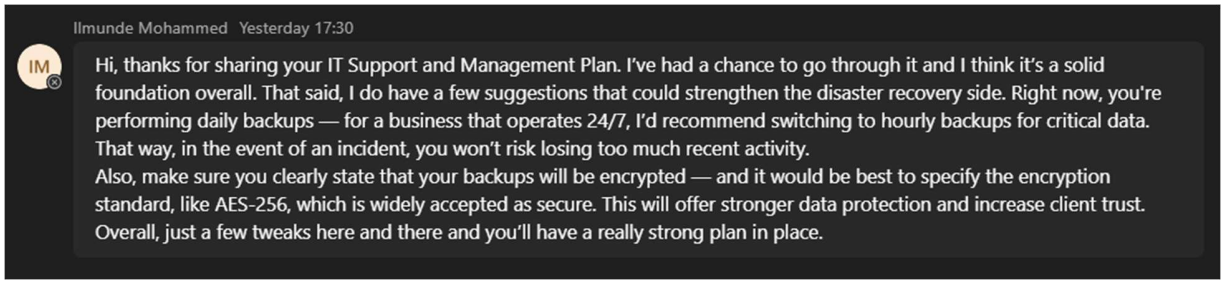
investment, while sustainability practices will demonstrate corporate responsibility and reduce costs with energy and waste. Preventive maintenance is a further protection step for the business which reduces the chance of unexpected IT failures. All these steps/actions together will ensure that the start-up business has a fit-for-purpose, and future ready IT environment.

Review and Refinement

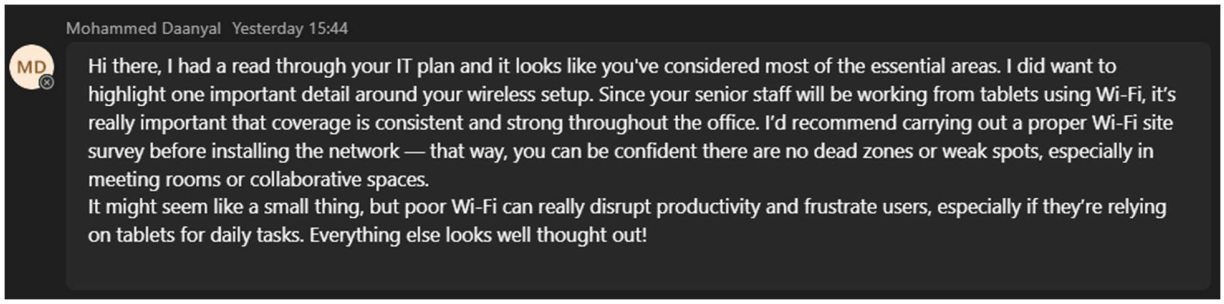
The original plan was reviewed with two colleagues: an IT technician and Business Manager. The IT technician who has experience managing small business IT infrastructures, suggested to enhance the backup strategy by including backup encryption to improve the security and increasing the backup frequency for critical data. While the Business Manager pointed mainly on the Wi-Fi infrastructure planning. Highlighting that tablet being used by senior staff needs to have the efficiencies coverage by conducting full Wi-Fi site survey before deployment to ensure complete signal coverage across the office space.

Based on this feedback, the plan was refined to include hourly backups for critical data instead of just daily backups. Backups were also specified to use AES-256 encryption to meet the industry standards for security. Additionally, a Wi-Fi coverage survey was added to the deployment phase to guarantee full signal strength across the entire building.

IT Technician



Business Manager



Evaluation of Refined Plan

The refinement was made based on the external feedback that was been suggested by an IT Technician and Business Manager, the feedback was significant to enhance the IT support and Management Plan, which ensures to meet the client's operational and business needs. The decision made was to increase the backup frequency and introduce encrypted storage strengthens the disaster recovery and data protection measures, addressing the client's need for system availability 24 hours a day, 365 days a year. Including a Wi-Fi survey during the network deployment to ensure that all users, particularly senior staff using tablets will be experience strong, uninterrupted wireless connectivity which supports the efficient working practices.

The plan is fit for purpose because it balances the technical resilience, scalability, security, and environmental responsibility. The incident strategies minimise downtime, disaster recovery plans protect data and maintain continuity and capacity planning allows future expansion without costly upgrades. Sustainability planning shows commitment to green practices, while preventive maintenance reduces the long-term risk of system failure.

Feedback from others was crucial in refining the plan. It led to more specific and stronger support decisions that not only improve technical quality but also demonstrate a deep understanding of the business's needs.

Although the refined IT support and Management Plan fully meets the client's requirements, there are some potential drawbacks to consider. The main limitation is the high initial cost of implementing the infrastructure which includes purchasing high-performance desktop PCs, professional-grade Wi-Fi equipment, secure cloud backups and helpdesk management systems. These upfront expenses could present financial challenges. Additionally, the plan introduces a level of complexity that will require skilled IT staff and outsourced services to manage backups, security maintenance schedules and incident responses effectively.

Another potential drawback is the company's dependence on external cloud service providers and reliable internet connectivity. Which any downtime from these providers will result in disruption business operations despite the system being designed for 24/7 availability. Overall, these limitations are manageable but should be acknowledged and monitored to ensure the IT system continues to deliver its intended benefits long-term.

Conclusion

Overall, the proposed solution of office layout and IT support and management plan for the company which covers incident response, disaster recovery, capacity planning, sustainability, and environmental and preventive maintenance, which will ensure to manage and monitor any threats or vulnerabilities that can affect the business operational.